```
class Shop
  #type string
  shopName
  #type Array<Item>
  items
  #type tuple(k)
  point

  function Shop(shopName, location,
items)                                          #1
  function order(item,
customer)                                                  #2
```

#1 Constructor for the class.
#2 Perform the actual purchase (details of this method can't be generalized, and they are not relevant to our discussion: just imagine this will update stock and start the process of actually shipping the goods to the customer).

The core of the problem, in this setting, is to keep an updated list of stores and retrieve the closest one to a customer, each time an order is placed on the website.

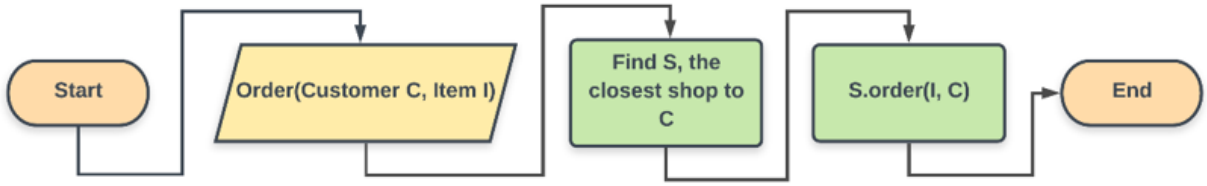


Figure 11.2 Simplified workflow for an order on the e-commerce site: after receiving the order from the customer, look up the closest retailer (assuming there will be always one selling the item ordered) and then place the order with it. Since we assume everything